

From Primitives to the Rate of Becoming

Slow Light, Stopped Light, Optical Lattice Clocks, and Gravitational Redshift from
Substrate Primitives

Allen Proxmire

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1 The Question

What this walkthrough derives

Three experimental programs sit at the frontier of atomic, molecular, and optical (AMO) physics:

1. **Hau (1999)**: light propagating through an ultracold atomic cloud under electromagnetically-induced-transparency (EIT) conditions has its group velocity reduced from $c \approx 3 \times 10^8$ m/s to ~ 17 m/s — a reduction by factor $\sim 10^7$.
2. **Hau (2001)**: in the same EIT system, adiabatic ramp of the control field to zero stores the optical pulse coherently in atomic ground-state coherence; subsequent reactivation retrieves the pulse with phase information preserved. Light is stopped, then released.
3. **Katori (2003) and Ye (et seq.)**: optical lattice clocks based on the $^1S_0 \leftrightarrow ^3P_0$ transition in alkaline-earth atoms (Sr, Yb) reach fractional frequency stabilities of $\sim 10^{-19}$, well past cesium-fountain microwave clocks. Networked clocks at millimeter-scale separation resolve gravitational redshift directly.

The walkthrough derives, from substrate primitives:

- The **substrate-level definition of local rate of becoming** as rate of phase accumulation in pre-individuation amplitudes.
- The **speed of light** c as the coarse-grained expression of this rate; substrate c is a fundamental constant.
- **Slow light** as group-velocity reduction of a dressed polariton, with substrate c unchanged.
- **Stopped light** as adiabatic identity transfer between channel types.
- **Optical lattice clocks** as direct probes of local becoming rate via phase accumulation.
- **Gravitational redshift** as ED-gradient curvature in the becoming rate.
- **Clock networks** as differential ED-gradient sensors.

The walkthrough is fully self-contained: every required EIT algebra, dark-state-polariton derivation, adiabatic-rotation step, magic-wavelength condition, and gravitational-redshift identity is derived inline.

What Event Density claims

All three results — slow light, stopped light, optical lattice clocks — are FORCED at the substrate level by a single foundational primitive: **the local rate of becoming**, defined precisely as the rate of phase accumulation in pre-individuation amplitudes. The mechanism is:

- Substrate c is the fundamental rate at which the rule-type structure propagates. It is a constant.
- A medium dresses the photon's participation rule with atomic-coherence rule structure, producing a polariton — a mixed-channel-type rule structure whose group velocity is the coarse-grained becoming rate of the dressed object, generally less than c .

- Adiabatic switching of the medium’s coupling rotates the polariton’s identity continuously between photonic and atomic-spin-coherence components.
- Atomic clock transitions are direct probes of the local becoming rate.
- Gravitational redshift is the ED-gradient curvature in the local becoming rate.
- Clock networks measure differential ED-gradient curvature with millimeter-scale resolution.

The substrate-level mechanism, in one sentence: **all three Hau–Katori–Ye phenomena manipulate or measure the local rate of becoming, which is the substrate-level rate of phase accumulation that coarse-grains to both the speed of light and the tick rate of atomic clocks.**

2 The Primitives

2.1 P-RB-1. Local rate of becoming

The *local rate of becoming* at a substrate-level point x is the rate at which a chain’s pre-individuation amplitude at x accumulates phase per unit substrate-tick. Operationally, for a chain whose pre-individuation amplitude is $\psi(x, t) = \sum_n c_n(t) e^{-i\theta_n(t)} |r_n(x)\rangle$ aligned with eigen-rule $|r_n(x)\rangle$ of eigen-energy $E_n(x)$, the becoming rate is

$$\omega_{\text{becoming}}(x) \equiv d\theta_n/dt = E_n(x)/\hbar.$$

Substrate-level meaning. The pre-individuation amplitude is the substrate-level superposition of consistent rule continuations weighted by complex amplitudes. Phase accumulation in this amplitude is the substrate-level analog of “passage of time” for the chain’s identity: each phase increment corresponds to one substrate-tick of becoming.

Two coarse-grained quantities. When coarse-grained to standard observables:

- For massless chains (light), the rate of becoming sets the propagation speed: c is the coarse-grained becoming rate of the substrate’s rule-type structure propagating freely.
- For massive chains, the rate of becoming is the standard QM angular frequency $\omega = E/\hbar$.

This is the load-bearing primitive of the walkthrough.

2.2 P-RB-2. Light-like chain

A *light-like chain* is a minimal ED-channel (multiplicity $M = 1$) whose participation rule has zero rest energy: $E = \hbar\omega = \hbar c|\mathbf{k}|$.

Substrate c is a constant. This is non-negotiable. Slow light, refractive index, gravitational time dilation, and other apparent modifications of light’s propagation rate are all *coarse-grained dressed-object* effects. The substrate-level photon always propagates at c .

2.3 P-RB-3. ED-gradient

An *ED-gradient* is a spatial variation in the local rate of becoming: $\nabla_x \omega_{\text{becoming}}(x) \neq 0$.

Sources include coupling to a medium, gravitational potential gradients, and spatially-varying field configurations. Higher-order spatial variation produces ED-gradient curvature.

2.4 P-RB-4. Identity alignment

The *identity alignment* of a chain at substrate-level position x is its commitment to one specific participation rule from the local alignment set $\mathcal{R}(x) = \{r_n(x)\}$. Smooth changes in the substrate’s rule structure can preserve a chain’s identity alignment if the change is slow enough (adiabatic).

2.5 P-RB-5. Medium-induced rule coupling

A *medium* is a substrate region containing additional chains whose rule-type structure couples to a propagating chain’s rule structure. When a photonic chain $\hat{a}$ at frequency ω couples to atomic chains in a 3-level Λ configuration with control field $\hat{\Omega}_c$, the effective interaction is

$$H_{\text{coupling}} = g(\hat{a}^\dagger \sigma_{eg} + \hat{a} \sigma_{ge}) + \Omega_c(\sigma_{em} + \sigma_{me}) + \text{h.c.}$$

This produces *polaritons* — mixed-channel-type rule structures, derived inline in §4.

2.6 P-RB-6. Coherence storage

Coherence storage is the substrate-level transfer of a chain’s identity alignment from one channel type (photonic) to another (atomic spin coherence) via adiabatic rotation of the participation-rule structure. The adiabatic condition: $|d/dt(\Omega_c, g)| \ll \Delta_{\text{gap}}$.

2.7 P-RB-7. Local becoming measurement

Local becoming measurement is any operational protocol that probes the local rate of becoming directly. The atomic-clock transition is the canonical example: a chain in superposition between two long-lived eigen-rules accumulates relative phase at rate $(E_2 - E_1)/\hbar$, which is the local becoming-rate difference between the two rules.

No additional substrate primitives are introduced beyond P-RB-1 through P-RB-7.

3 Deriving the Speed of Light from the Rate of Becoming

3.1 Substrate c as V1-kernel propagation rate

The substrate’s V1 kernel encodes how rule-type structure propagates between adjacent substrate points per substrate-tick. The propagation rate is a substrate-level constant: in the coarse-grained limit, the V1 kernel reduces to a $\delta(x - ct)$ propagation pattern.

Standard physics identifies this propagation rate with the speed of light $c \approx 3 \times 10^8$ m/s. The substrate-level statement is: c is the V1-kernel propagation rate, and it is invariant under all coordinate, gauge, and material transformations.

3.2 Group velocity vs phase velocity

For a chain propagating freely in vacuum, both phase velocity ($v_\phi = \omega/|\mathbf{k}|$) and group velocity ($v_g = d\omega/d|\mathbf{k}|$) equal c . For a chain propagating in a medium, the dispersion relation is modified. The chain’s coarse-grained becoming rate becomes a non-linear function $\omega(|\mathbf{k}|)$, and the group velocity $v_g = d\omega/d|\mathbf{k}|$ can be much less than c . **The substrate c is unchanged**; the modification lives entirely in the dispersion relation of the dressed object.

3.3 Refractive index from ED-gradient slope

The refractive index n is defined operationally by $n(\omega) = c/v_{\text{phase}}(\omega)$. In the substrate-level picture, n is the coarse-grained image of an ED-gradient slope.

Substrate-level statement. Refractive index is not a property of an underlying medium that “slows” light. It is the coarse-grained image of how the dressed-object’s rule structure couples the photonic channel to the medium’s atomic-coherence channel.

3.4 Why substrate c must be invariant

If c varied with environment, the substrate's V1-kernel propagation would not be a single rule-type primitive — it would be a family of rules. This would violate the substrate's primitive economy: the kernel would no longer be one primitive.

4 Slow Light as Mixed-Channel Rule Structure (Hau 1999)

4.1 Setup: the 3-level Λ system

Hau's slow-light experiment uses ultracold Na atoms with three relevant atomic levels. A weak probe field at frequency ω_p couples $|g\rangle \leftrightarrow |e\rangle$. A strong control field at frequency ω_c couples $|m\rangle \leftrightarrow |e\rangle$. Two-photon resonance: $\omega_p - \omega_c$ matches the $|g\rangle \rightarrow |m\rangle$ ground-state splitting.

4.2 Three-level Hamiltonian

In the rotating-wave approximation:

$$H_{\text{atom}} = -\Delta_p|e\rangle\langle e| + g\hat{a}|e\rangle\langle g| + \Omega_c|e\rangle\langle m| + \text{h.c.}$$

For N atoms, the collective coupling to the photon mode is $g\sqrt{N}$.

4.3 Dark state and EIT

The atomic Hamiltonian admits a *dark state* — an eigenstate that does not couple to the excited level:

$$|D\rangle = \cos\theta|g\rangle - \sin\theta|m\rangle, \quad \tan\theta = g\sqrt{N}\hat{a}/\Omega_c.$$

Verification: the matrix element of H_{atom} between $|D\rangle$ and $|e\rangle$ is

$$\langle e|H_{\text{atom}}|D\rangle = g\sqrt{N}\hat{a}\cos\theta - \Omega_c\sin\theta = 0$$

for the given $\tan\theta$. The dark state is transparent to the resonant absorption process. This is the EIT condition.

4.4 Dark-state polariton (mixed-channel rule structure)

The full system admits an eigenstate that is a coherent superposition of photonic and atomic-coherence components:

$$|\Psi_{\text{polariton}}\rangle = \cos\theta|1_{\text{photon}}, g\rangle - \sin\theta|0_{\text{photon}}, m\text{-coherence}\rangle,$$

with mixing angle $\tan\theta = g\sqrt{N}/\Omega_c$.

Substrate-level meaning. The polariton is one substrate object whose rule structure is composite. It is not a photon traveling alongside an atomic excitation; it is one chain whose identity alignment spans two channel types.

4.5 Group velocity of the polariton

The polariton's effective Hamiltonian at two-photon resonance is

$$H_{\text{polariton}} = ck\cos^2\theta.$$

The group velocity is

$$v_g = dE_{\text{polariton}}/d(\hbar k) = c\cos^2\theta = c \cdot \frac{\Omega_c^2}{\Omega_c^2 + g^2N}.$$

For weak control field ($\Omega_c \ll g\sqrt{N}$): $v_g \rightarrow c \cdot \Omega_c^2/(g^2 N) \ll c$.

For typical Hau-experiment values ($g\sqrt{N} \sim 10$ GHz, $\Omega_c \sim 10$ MHz): $v_g \sim 10\text{--}30$ m/s. **Hau measured** ~ 17 m/s.

4.6 Substrate-level reading

- The propagating object is a polariton, not a substrate-level photon.
- The polariton’s group velocity is the coarse-grained becoming rate of the dressed mode.
- When the photonic component dominates ($\cos^2 \theta \approx 1$), the polariton becomes light-like.
- When the atomic component dominates ($\sin^2 \theta \approx 1$), the polariton becomes essentially stationary.
- **Substrate c is unchanged.**

5 Stopped Light as Identity Transfer (Hau 2001)

5.1 Adiabatic ramp of the control field

Slowly reduce the control-field Rabi frequency: $\Omega_c(t) \rightarrow 0$ over a timescale long compared to the gap between the dark-state polariton and the excited atomic states. The mixing angle evolves: $\tan \theta(t) = g\sqrt{N}/\Omega_c(t) \rightarrow \infty$, so $\theta(t) \rightarrow \pi/2$. The polariton rotates from photonic-dominated to atomic-dominated:

$$|\Psi_{\text{polariton}}\rangle \rightarrow -|0_{\text{photon}}, m\text{-coherence}\rangle \text{ as } \Omega_c \rightarrow 0.$$

5.2 Adiabatic identity preservation

Provided the ramp is slow enough, the chain’s identity alignment is preserved. **The chain’s identity is now stored entirely in the medium’s rule structure.** No photon is propagating; the chain’s identity exists as ground-state atomic coherence.

5.3 Storage time and decoherence

The storage time is limited by the lifetime of the atomic ground-state coherence — typically much longer than the excited-state lifetime. In Hau’s experiment, storage times of ~ 1 ms were demonstrated.

5.4 Retrieval via reversed ramp

Re-applying the control field reverses the polariton rotation. The chain’s identity returns to a photonic-dominated polariton, which propagates out of the medium with the original frequency, polarization, and phase information preserved.

Substrate-level reading. Stopped light is adiabatic identity transfer between channel types. No state is “frozen”; no photon is “captured.” The substrate-level mechanism is continuous identity transfer, parameterized by the control-field strength.

6 Optical Lattice Clocks as Becoming-Rate Probes (Katori, Ye)

6.1 The clock transition

Strontium-87 has a metastable 3P_0 state above the 1S_0 ground state. The transition $^1S_0 \leftrightarrow ^3P_0$ at 698 nm has natural linewidth ~ 1 mHz.

The chain’s pre-individuation amplitude in the clock’s superposition is

$$\psi_{\text{clock}}(t) = \frac{1}{\sqrt{2}}[|^1S_0\rangle + e^{-i\omega_{\text{clock}}t}|^3P_0\rangle],$$

where $\omega_{\text{clock}} = (E_{^3P_0} - E_{^1S_0})/\hbar \approx 2\pi \cdot 4.3 \times 10^{14}$ Hz.

6.2 Clock observable as becoming-rate measurement

The clock’s operational observable is the rate at which the relative phase between the two state components accumulates:

$$d\theta_{\text{relative}}/dt = \omega_{\text{clock}} = (E_{^3P_0} - E_{^1S_0})/\hbar.$$

By P-RB-1, the clock observable *is* the local becoming rate of the clock-state participation rule, made operationally accessible via interferometry between the two state components.

The clock measures local rate of becoming directly.

6.3 Magic wavelength as substrate-level Stark cancellation

When the clock atoms are trapped in an optical lattice, the lattice light induces AC Stark shifts on both clock states:

$$\Delta\omega_{\text{clock}}(\lambda) = -\frac{I}{2\hbar}[\alpha_{^3P_0}(\lambda) - \alpha_{^1S_0}(\lambda)].$$

At the *magic wavelength* $\lambda_m \approx 813$ nm for Sr, $\alpha_{^3P_0}(\lambda_m) = \alpha_{^1S_0}(\lambda_m)$, so $\Delta\omega_{\text{clock}}(\lambda_m) = 0$.

Substrate-level statement. At λ_m , the substrate-level photon-atom rule-type coupling produces *no net rule-type-energy shift* on the clock-transition rule structure.

6.4 Lamb-Dicke regime as motional-sideband suppression

When the trapping potential is strong enough that $\eta \equiv k_{\text{clock}} \cdot a_{\text{vib}} \ll 1$ (Lamb-Dicke parameter), the atomic motion does not modulate the clock-transition phase accumulation. Doppler shifts and motional sidebands are suppressed.

6.5 Clock precision as substrate-level isolation

The optical lattice clock’s fractional frequency stability is

$$\sigma_y(\tau) \sim 1/(\omega_{\text{clock}}\sqrt{N \cdot T_{\text{coh}} \cdot \tau}).$$

The current frontier is $\sigma_y \sim 10^{-19}$ over hour-scale integration.

6.6 Why clocks measure geometry of becoming, not “time”

In the standard reading, an atomic clock measures time. In the substrate-level reading, the clock measures *the local rate of becoming* — the rate of phase accumulation in the chain’s pre-individuation amplitude at the clock’s spatial position. **The substrate-level claim:** what optical lattice clocks measure with 10^{-19} precision is not a universal “time” but a *local geometric quantity*.

7 Gravitational Redshift as ED-Gradient Curvature

7.1 Gravitational potential as ED-gradient

In the weak-field limit, the gravitational potential $\Phi(\mathbf{x})$ is the coarse-grained image of an ED-gradient in the local rate of becoming:

$$\omega_{\text{becoming}}(\mathbf{x}) = \omega_0[1 + \Phi(\mathbf{x})/c^2],$$

where ω_0 is the becoming rate at infinite distance from any mass.

7.2 Local becoming rate $d\tau$

A clock at position $\mathbf{x}$ measures

$$\omega_{\text{becoming}}(\mathbf{x}) \cdot dt = (E/\hbar)(1 + \Phi/c^2) dt = (E/\hbar) d\tau,$$

where $d\tau = (1 + \Phi/c^2) dt$ is the *proper-time interval*. In the weak-field limit:

$$d\tau \approx \sqrt{1 + 2\Phi/c^2} dt = \sqrt{g_{00}} dt,$$

matching the standard general-relativistic identity.

7.3 Redshift between two heights

Two clocks at heights h_1 and h_2 have local potentials $\Phi_i = gh_i$. Their becoming-rate ratio is

$$\omega_1/\omega_2 \approx 1 + g(h_1 - h_2)/c^2.$$

For $h_1 > h_2$: $\omega_1 > \omega_2$, so the upper clock ticks faster:

$$\Delta\omega/\omega = -g\Delta h/c^2 \quad (\text{gravitational redshift formula}).$$

7.4 Substrate-level reading

Gravitational redshift is not a property of light; it is a property of the substrate's becoming-rate gradient. The redshift is FORCED by the ED-gradient in $\omega_{\text{becoming}}(\mathbf{x})$.

Optical lattice clocks detect ED-curvature directly. For $\Delta h = 1$ mm and Earth's $g \approx 10$ m/s²: $\Delta\omega/\omega \approx 10^{-19}$. This is the precision frontier of current clocks.

7.5 Why GR identifies g_{00} with becoming-rate ratio

The ED-substrate-gravity reading: g_{00} is the coarse-grained image of $\omega_{\text{becoming}}/\omega_0$, the ratio of local becoming rate to the reference becoming rate at infinite distance.

8 Clock Networks as ED-Gradient Sensors

8.1 Differential clock measurements

A network of optical lattice clocks at different spatial positions can be compared via fiber-optic frequency dissemination. The differential measurement

$$\Delta\nu_{ij} = \nu_i - \nu_j$$

cancels common-mode noise and isolates the *position-dependent* becoming-rate difference.

8.2 Sensitivity to ED-gradients

For sufficiently isolated clocks, the differential signal becomes dominated by the gravitational potential difference $\Delta\Phi/c^2$. Current state-of-the-art Sr clock networks can resolve potential differences corresponding to ~ 1 mm altitude differences over hour-scale integration.

8.3 Substrate-level mapping

Each clock measures its local becoming rate. A network of clocks samples ω_{becoming} at many points; differential measurements reveal the spatial structure — the substrate’s ED-gradient geometry. Clock networks are *ED-geometry sensors*.

8.4 Why this is more than gravimetry

Standard gravimeters measure $g(\mathbf{x})$. Clock networks measure gravitational *potential* $\Phi(\mathbf{x})$ directly via the becoming-rate gradient. Clock networks have access to the more fundamental substrate-level field.

8.5 Future tests

- **Variations of fundamental constants.**
- **Dark-matter searches** via oscillations in fundamental constants.
- **Gravitational-wave detection** as transient signals in differential measurements.

9 Substrate-Level Reading

Standard AMO/GR object	Substrate-level meaning
Speed of light c	Coarse-grained becoming rate of free-photon rule structure
Refractive index $n(\omega)$	ED-gradient slope of dressed becoming-rate function
Group velocity v_g	Coarse-grained becoming rate of the dressed (polariton) chain
Slow light	Polariton becoming-rate reduction via mixing with atomic-coherence
Stopped light	Identity adiabatically transferred from photonic to atomic-coherence channel
Atomic transition frequency	Local becoming rate; coarse-grains to $E/\hbar$
Magic wavelength	Substrate-level Stark cancellation on clock-transition rule structure
Lamb-Dicke regime	Decoupling of internal-state from external-motion rule structure
Optical lattice clock	Direct probe of local rate of becoming
Gravitational potential $\Phi(\mathbf{x})$	Coarse-grained ED-gradient in local becoming rate
Gravitational redshift	Ratio of becoming rates at two points
Time dilation $d\tau = \sqrt{g_{00}} dt$	Local becoming-rate ratio relative to reference
Clock network	Spatially-distributed sensor of ED-geometry

9.1 The unified phenomenon

The Hau, Katori, and Ye experimental programs are a single substrate-level enterprise: they manipulate or measure the local rate of becoming. The substrate-level mechanism is the same in all three: the local rate of becoming is the foundational substrate primitive; light propagation, atomic clock ticking, and gravitational time dilation are different coarse-grained expressions of the same underlying becoming-rate field.

10 What's Forced, What's Inherited, What's Open

What is FORCED by the substrate ontology

- Local rate of becoming as rate of phase accumulation in pre-individuation amplitudes (P-RB-1).
- Speed of light c as constant V1-kernel propagation rate.
- Group velocity reduction in slow light via mixed-channel rule structure.
- Stopped light as adiabatic identity transfer.
- Optical-clock observable as local rate of becoming via P-RB-1.
- Magic-wavelength condition as substrate-level Stark cancellation.
- Gravitational redshift as ED-gradient curvature.
- Identity of GR's g_{00} with becoming-rate ratio.

What is FORM-FORCED-INHERITED (and re-derived inside this document)

Three-level Λ -system Hamiltonian in rotating-wave approximation; dark-state structure in EIT; dark-state polariton composition; group velocity formula; AC Stark shift formula; Lamb-Dicke parameter; adiabatic-evolution identity preservation; weak-field metric expansion; Lindblad master-equation framework.

What remains OPEN

- Many-body ED-flow beyond mean-field treatment.
- Nonlinear becoming gradients (hyperpolarizability shifts).
- Relativistic corrections to becoming rate in strong-field regimes.
- Quantum-coherence-enhanced clocks reaching Heisenberg-limited precision.
- Frequency-comb-based clock comparisons.
- Drift of fundamental constants.
- Tests of Lorentz/CPT invariance.

11 What This Argument Establishes

Claim 1. The local rate of becoming, defined as the rate of phase accumulation in pre-individuation amplitudes, is a foundational substrate primitive (P-RB-1). It coarse-grains to standard QM angular frequency $E/\hbar$ for atomic-state superpositions and to substrate c for free-photon propagation.

Claim 2. The substrate-level c is a constant — the V1-kernel propagation rate. Slow light, refractive index, and other apparent modifications are coarse-grained dressed-object effects; substrate c is invariant.

Claim 3. Slow light (Hau 1999) is FORCED as group-velocity reduction of the dark-state polariton, with photon-component weight $\cos^2 \theta = \Omega_c^2 / (\Omega_c^2 + g^2 N)$. Group velocity $v_g = c \cos^2 \theta$.

Claim 4. Stopped light (Hau 2001) is FORCED as adiabatic identity transfer from photonic to atomic-coherence channel as $\Omega_c \rightarrow 0$. Retrieval is the reverse rotation; no information is lost.

Claim 5. Optical lattice clocks (Katori, Ye) measure the local rate of becoming directly via interferometry. Magic wavelength is the substrate-level condition for zero net rule-type-energy shift; Lamb-Dicke regime decouples internal-state from motional rule structure.

Claim 6. Gravitational redshift is FORCED by the ED-gradient in local becoming rate: $\omega_{\text{becoming}}(\mathbf{x}) = \omega_0(1 + \Phi(\mathbf{x})/c^2)$. The GR identity $d\tau = \sqrt{g_{00}} dt$ is the standard-physics statement of the substrate-level becoming-rate ratio.

Claim 7. Clock networks are ED-geometry sensors. Differential measurements at 10^{-19} precision resolve millimeter-scale ED-gradient features on Earth.

Claim 8 (negative). No new substrate primitives are required beyond the seven listed in §2.

Claim 9 (scope-limit). This walkthrough does not derive: many-body ED-flow beyond mean-field; nonlinear becoming gradients; strong-field relativistic corrections; quantum-coherence-enhanced clock precision; frequency-comb-based comparison; fundamental-constant drift; Lorentz/CPT-violation tests.

The unified statement. The Hau–Katori–Ye cluster is one substrate-level enterprise: manipulate or measure the local rate of becoming. The local rate of becoming is the rate of phase accumulation in pre-individuation amplitudes — the substrate’s foundational rate primitive. Speed of light, atomic-clock tick rate, refractive index, gravitational time dilation, and clock-network gradient sensing are all coarse-grained expressions of this same becoming-rate field. The substrate-level mechanism unifies AMO physics with general relativity at the level of the becoming-rate field.

References

- Hau, L. V., Harris, S. E., Dutton, Z., Behroozi, C. H. *Light speed reduction to 17 metres per second in an ultracold atomic gas.* Nature **397**, 594 (1999).
- Liu, C., Dutton, Z., Behroozi, C. H., Hau, L. V. *Observation of coherent optical information storage in an atomic medium using halted light pulses.* Nature **409**, 490 (2001).
- Katori, H. *Spectroscopy of Strontium Atoms in the Lamb-Dicke Confinement.* Phys. Rev. Lett. **91**, 173005 (2003).
- Ye, J., Kimble, H. J., Katori, H. *Quantum state engineering and precision metrology using state-insensitive light traps.* Science **320**, 1734 (2008).
- Bloom, B. J. et al. *An optical lattice clock with accuracy and stability at the 10^{-18} level.* Nature **506**, 71 (2014).
- Bothwell, T. et al. *Resolving the gravitational redshift across a millimetre-scale atomic sample.* Nature **602**, 420 (2022).
- Fleischhauer, M., Imamoglu, A., Marangos, J. P. *Electromagnetically induced transparency: Optics in coherent media.* Rev. Mod. Phys. **77**, 633 (2005).
- Lukin, M. D. *Trapping and manipulating photon states in atomic ensembles.* Rev. Mod. Phys. **75**, 457 (2003).
- Ludlow, A. D., Boyd, M. M., Ye, J., Peik, E., Schmidt, P. O. *Optical atomic clocks.* Rev. Mod. Phys. **87**, 637 (2015).
- Misner, C. W., Thorne, K. S., Wheeler, J. A. *Gravitation.* W. H. Freeman (1973).